

```

list=[2,7,4,1,3,6] #count pairs whose sum equals to 10
count=0
for i in range(len(list)):
    for j in range(i+1,len(list)):
        if list[i]+list[j]==10:
            count+=1
print(count)

```

Output:2

```

#find range of a list
array=[5,3,8,1,0,4]
if len(array)<3:
    print("range is not possible")
else:
    max=max(array)
    min=min(array)
    range_value=max-min
    print(range_value)

```

Output: 8

```

#Return A raised to power m
A = [
    [1, 2],
    [3, 4]
]

m = 2

result = A

for _ in range(m - 1):
    temp = [[0, 0], [0, 0]]

    for i in range(2):
        for j in range(2):
            for k in range(2):
                temp[i][j] += result[i][k] * result[k][j]

    result = temp
print(result)

```

Output: [[7, 10], [15, 22]]

```

#highest occurring character
string="hippopotamus"
max_count=0
max_char= ''
for char in string:
    count=string.count(char)
    if count>max_count:
        max_count=count
        max_char=char
print(max_char)
print(max_count)

```

Output: p,3

```

#mean,median,mode
import random
import statistics

nums = [random.randint(1, 10) for _ in range(25)]

print("Numbers:", nums)

mean = statistics.mean(nums)
median = statistics.median(nums)
mode = statistics.mode(nums)

print("Mean =", mean)
print("Median =", median)
print("Mode =", mode)

```

Output: Numbers: [9, 7, 9, 2, 10, 6, 9, 6, 5, 4, 8, 4, 3, 1, 9, 1, 1, 7, 5, 4, 6, 8, 2, 3, 5]

Mean = 5.36

Median = 5

Mode = 9